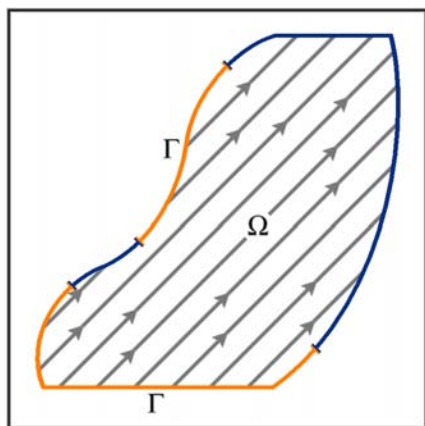


18.336 Numerical Methods for Partial Differential Equations

Fundamental Concepts

Domain $\Omega \subset \mathbb{R}^n$ with boundary $\partial\Omega$

$$\left\{ \begin{array}{ll} \text{PDE} & \text{in } \Omega \\ \text{b.c.} & \text{on } \Gamma \subset \partial\Omega \end{array} \right\}$$



PDE = “partial differential equation”

b.c. = “boundary conditions”

(if time involved, also i.c. = “initial conditions”)

Def.: An expression of the form

$$F(D^k u(x), D^{k-1} u(x), \dots, Du(x), u(x), x) = 0, \quad x \in \Omega \subset \mathbb{R}^n \quad (1)$$

is called k^{th} order PDE,

where $F : \mathbb{R}^k \times \mathbb{R}^{n^{k-1}} \times \dots \times \mathbb{R}^n \times \mathbb{R} \times \Omega \rightarrow \mathbb{R}$ is given,

and $u : \Omega \rightarrow \mathbb{R}$ is the unknown.

A function u satisfying (1) is called *solution* of the PDE.

$Du = (u_{x_1}, \dots, u_{x_n})$ $D^2u = \begin{pmatrix} u_{x_1 x_1} & \cdots & u_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ u_{x_n x_1} & \cdots & u_{x_n x_n} \end{pmatrix}$ $\vdots$	gradient (vector) Hessian (matrix) $\vdots$ etc.
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Def.: The PDE (1) is called...

(i) linear, if

$$\sum_{|\alpha| \leq k} a_\alpha(x) D^\alpha u = f(x)$$

homogeneous, if $f = 0$

(ii) semilinear, if

$$\sum_{|\alpha| \leq k} a_\alpha(x) D^\alpha u + F_0(D^{k-1}u, \dots, Du, u, x) = 0$$

(iii) quasilinear, if

$$\sum_{|\alpha| \leq k} a_\alpha(D^{k-1}u, \dots, Du, u, x) \cdot D^\alpha u + F_0(D^{k-1}u, \dots, Du, u, x) = 0$$

(iv) fully nonlinear, if neither (i), (ii) nor (iii).

Def.: An expression of the form

$$F(D^k u(x), D^{k-1} u(x), \dots, Du(x), u(x), x) = 0, \quad x \in \Omega \subset \mathbb{R}^n$$

is called k^{th} order system of PDE,

where $F : \mathbb{R}^{mn^k} \times \mathbb{R}^{mn^{k-1}} \times \dots \times \mathbb{R}^{mn} \times \mathbb{R}^m \times \Omega \rightarrow \mathbb{R}^m$

and $u : \Omega \rightarrow \mathbb{R}^m$, $u = (u^1, \dots, u^m)$.

Typically: # equations = # unknowns , i.e. $n = m$.

Some examples:

$u_t + u_x = 0$ linear advection equation

$u_t = u_{xx}$ heat equation

$u_{xx} = f(x)$ Poisson equation (1D)

$\nabla^2 u = f$ Poisson equation (nD)

$u_t + cu_x = Du_{xx}$ convection diffusion equation

$u_t + (\frac{1}{2}u^2)_x = 0 \Leftrightarrow u_t + u u_x = 0$ Burgers' equation (quasilinear)

$\nabla^2 u = u^2$ a semilinear PDE

$u_{tt} = u_{xx}$ wave equation (1D)

$\begin{pmatrix} u \\ v \end{pmatrix}_t = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} u \\ v \end{pmatrix}_x$ wave equation, written as a system

$$u_{tt} = v_{xt} = v_{tx} = u_{xx}$$

$u_t + uu_x = \epsilon u_{xxx}$ Korteweg-de-Vries equation

$\left\{ \begin{array}{l} \vec{u}_t + (\vec{u} \cdot \nabla) \vec{u} = -\nabla p + \nu \nabla^2 \vec{u} \\ \nabla \cdot \vec{u} = 0 \end{array} \right\}$ incompressible Navier-Stokes equation

[dynamic-algebraic system]

$\left\{ \begin{array}{l} h_t + (uh)_x = 0 \\ u_t + uu_x = -gh_x \end{array} \right\}$ shallow water equations

[system of hyperbolic conservation laws]

$|\nabla u| = 1$ Eikonal equation (nonlinear)

Why is the advection equation "transport"?

Consider $u_t + cu_x = 0$

Suppose $c=1$:

$u_t + u_x = 0$

The solution is

$u(x,t) = u_0(x-t)$

where u_0 is the initial shape.

The entire shape moved to the right. No change in height, width, area.

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fully nonlinear
now consider $|\nabla u| = 1$
In 2D:
 $\sqrt{u^2_x + u^2_y} = 1$
Notice:
 u^2_x (the derivative) is squared so the higher-order derivative don't appear linearly.

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For numerical PDEs, the single most important fact is:

$$\Delta u = \nabla \cdot (\nabla u) = u_{xx} + u_{yy} \quad (2D)$$

and it measures how far a point is from being the average of its neighbors, which is why it naturally appears in diffusion, heat flow, electrostatics, and Poisson equations.